Fall 2025

The George Washington University - Libraries and Academic Innovation

Introduction to Mapping in QGIS

bit.ly/qgis-workshop-docs

Image Description:

The logo for QGIS, a green Q with orange and yellow accents.

Roadmap

- **What is QGIS?**

- **Building a Map**
  - **Polygons**
  - **Points**
  - **Lines**

- **Styling your Map**

What is QGIS?

What is QGIS?

- QGIS is an open source application for GIS, or a Geographic Information System, which is used for displaying and analyzing spatial information (making maps).

- QGIS has very similar functionality to a program called ArcGIS, so most concepts you learn in one system are applicable to the other.

Building a Map

QGIS Plugins

- Plugins offer quality of life improvements and advanced functionality for QGIS.

- Installing plugins:
  - Click on “Plugins” on the menu bar
  - Then “Manage and Install Plugins”

- The plugin we’ll use is called “QuickMapServices”

Demo this - show a few different options, probably ESRI and Google.

Installing QuickMapServices

- After installed, find the “QuickMapServices” icon on the toolbar (blue globe with a green plus sign) and navigate to “Settings”

- With the settings window open, click the “More services” tab and then click on “Get contributed pack”

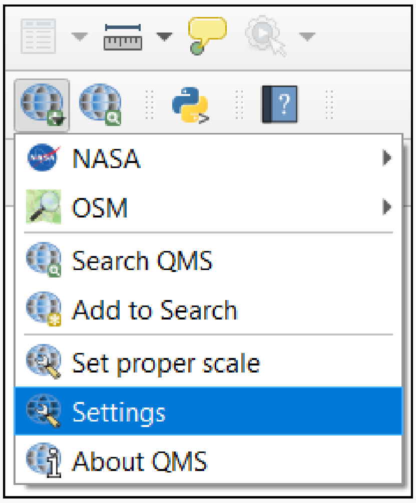


Image Description:

The "QuickMapServices" menu expanded, with the "Settings" option highlighted.

Adding a Base Layer

- Click on the “QuickMapServices” icon again, and you should see a list of “base layer” options that you can add to your map.

- Different base layers show different things! You’ll want to experiment depending on the information you want to show.

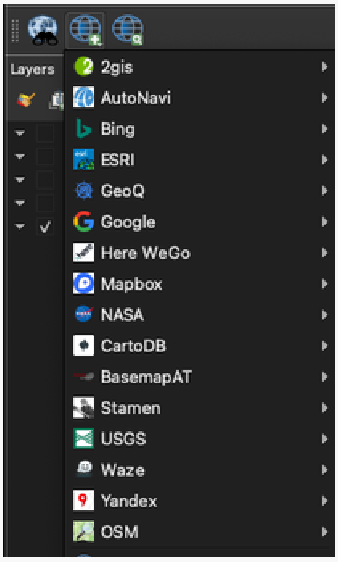
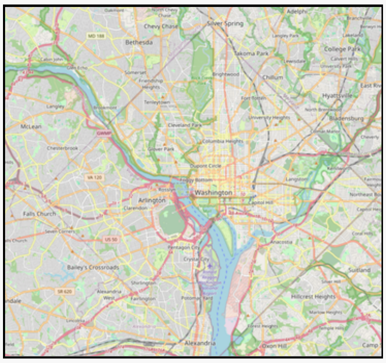


Image Description:

A screenshot of the dropdown menu of base map options in QuickMapServices.

Base Layer Examples



OSM Standard

Google Satellite

ESRI Gray (light)

Image description:

On the left, a map of DC with highlighted roads, parks, and water, labeled "OSM Standard".

In the middle, a satellite view of DC, appearing mostly grey and dark green, labeled "Google Satellite".

On the right, a very simple grey map of DC with no labels on the map. Labeled "ESRI Gray (light)"

Finding Geospatial Data

Polygons, Points, and Lines

Polygons

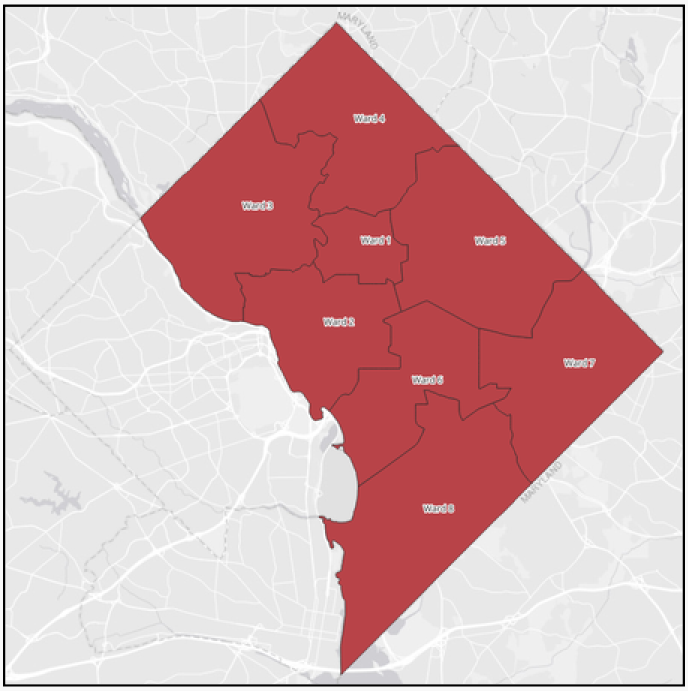


Image Description:

A light gray map of DC with polygons for wards overlaid. Wards are filled in red and have black borders.

Shapefiles (.shp)

- Specific file type formatted for compatibility with GIS software (like QGIS)
- The US Census publishes shapefiles for governmental divisions - states, census tracts, school districts, congressional districts, etc.
- Search for “Census Tigerline Shapefiles”

Demo downloading and importing shapefiles into QGIS.

DC Ward Shapefiles

- You won't see an option in the census dropdown menu for "Wards" - because DC isn't a state, finding DC data in the census can be a challenge.
- DC ward shapefiles can be found under: "**State Legislative District - Upper Chamber**"
- Once we download this file, we can drag and drop it into QGIS to import it.

Demo downloading and importing DC ward shapefile into QGIS.

Other DC Shapefiles

- DC publishes data, including shapefiles, for a broad range of topics on **https://opendata.dc.gov/**
- Another DC-specific political division is the system of **Advisory Neighborhood Commissions** and **Single Member Districts**.
- We can search for these shapefiles on OpenData DC:
    - **“Single Member District from 2023"**
    - **“Advisory Neighborhood Commissions from 2023"**

Demo downloading and importing DC ward shapefile into QGIS.

Points

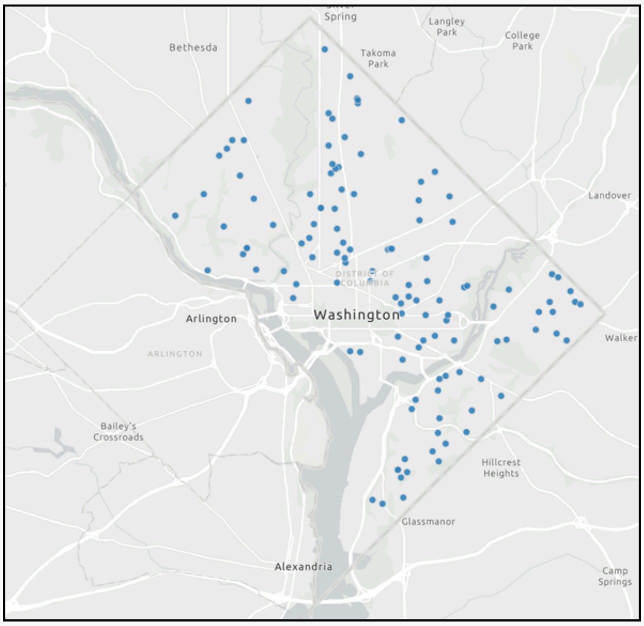


Image Description:

A light gray map of DC with blue dots scattered throughout, representing locations of DC public schools.

Importing Point Data

- Like polygons, points can also be stored as shapefiles!
- OpenData DC can be a great source of point shapefiles for a broad range of topics.
- Let's look at importing some point data!
  - Search for "**DC Public Schools**"

Demonstrate downloading DC Metro stations shapefile from opendatadc, drag into QGIS map

Creating Point Data

- Unlike polygons, it is relatively easy to create your own point data.
- CSV file
  - Required data:
    - X column (longitude)
    - Y column (latitude)
  - Optional data:
    - Columns for any other associated data

Latitude	Longitude	Name	Enrollment
38.899234	-77.048386	GWU	27814
38.938111	-77.088986	AU	14318
38.944712	-77.065584	UDC	4202
38.907613	-77.072248	Georgetown	20984

Don't actually demonstrate, just mention this

Importing Created Point Data

- Click on the “Layer” menu, then under “Add Layer”, then select “Add Delimited Text Layer”

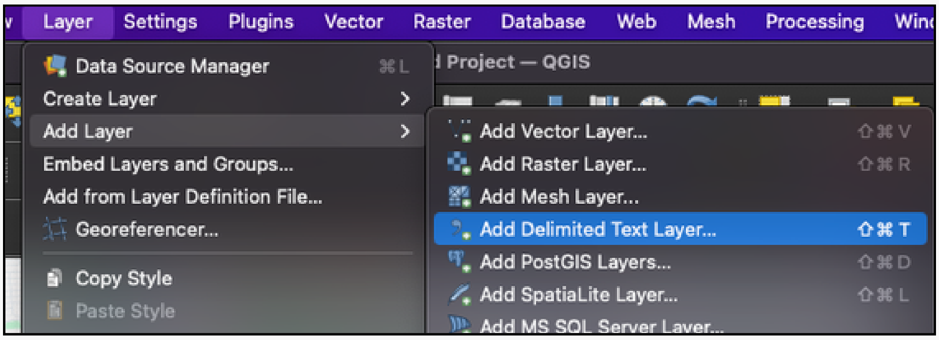


Image Description:

The expanded "Layer" menu, then the expanded "Add Layer" menu, and the highlighted option "Add Delimited Text Layer..."

Importing Created Point Data

- For “File name”, navigate to the CSV file with your data.
- Under “Geometry Definition” make sure your X field corresponds to your “longitude” column and Y field to “latitude”
- If you are using latitude/longitude, you may need to uncheck “DMS coordinates”

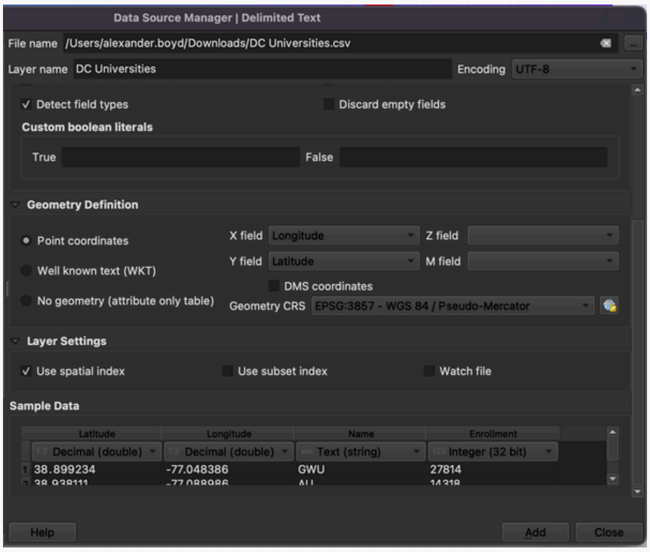


Image description:

A screenshot of the "Data Source Manager | Delimited Text" import options.

Lines

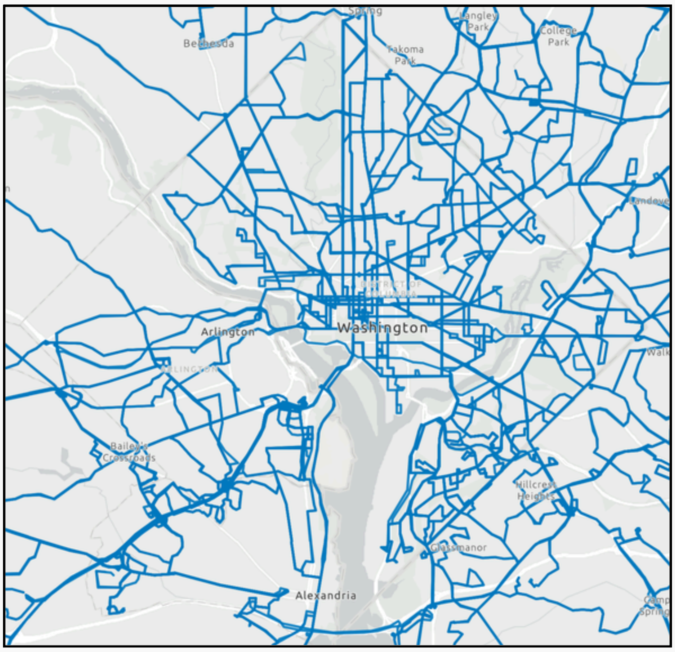


Image Description:

A light gray map of DC with blue lines drawn over many roads, representing Metro Bus routes.

Importing Line Data

- Like polygons and points, lines can also be stored as shapefiles! Line data is often used for roads, railways, busses, etc.

- Once again - OpenData DC and the census can be great resources for finding line files.

- Let's look at importing some line data!
  - Search for "**Metro Bus Lines**" on OpenData DC

Demonstrate downloading DC Metro stations shapefile from opendatadc, drag into QGIS map

Basic Analysis

Toolbox

- Click on the “Processing” tab and check “Toolbox”, or click on the light blue gear icon on the toolbar.
- This opens the “toolbox” panel, which includes a large number of pre-built tools for more in depth processing or analysis of your data

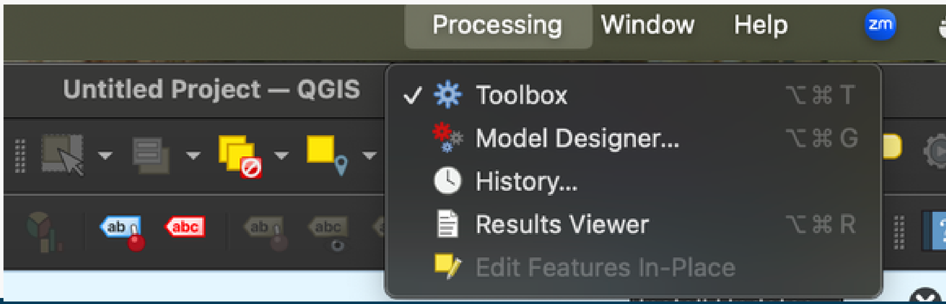


Image Description:

The menu that appears when right clicking a layer, with the "Properties..." button highlighted.

Example - Count Points in Polygon

- In the toolbox panel, search for “Count points in polygon” and click on the option that appears.
- This will open the tool, with explanation of how to run it, what inputs are needed, and what the results indicate.

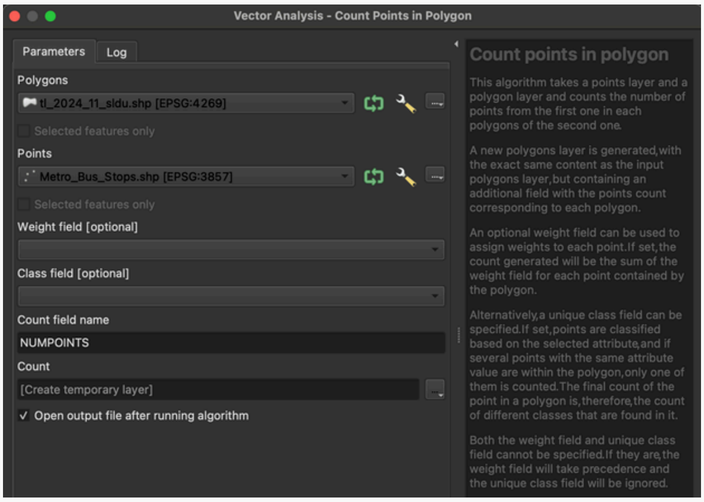
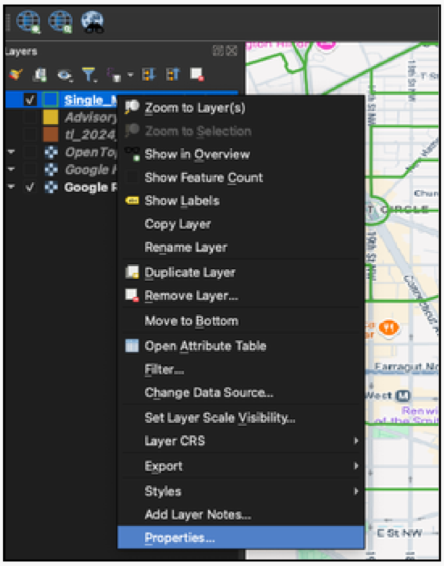


Image Description:

The menu that appears when right clicking a layer, with the "Properties..." button highlighted.

Styling Your Map

- In the “Layers” panel, right click on a layer and navigate to the “Properties” option.
- The two options we’ll take a look at are “Symbology” and “Labels”



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The menu that appears when right clicking a layer, with the "Properties..." button highlighted.

Questions?

Contact: alex.boyd@gwu.edu

Coding Consultations: https://library.gwu.edu/program-and-code

Data Consultations: https://library.gwu.edu/work-data